

7 3 3 4

(a) 3 1 1 1

$$\Omega = 7 \times 6 = 42$$

$$W = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 6$$

$$b = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 12$$

$$\therefore \frac{42 - (6 + 12)}{42} = \frac{\cancel{24}^{12} \cancel{4}^4}{\cancel{42}^{21} \cancel{7}^7} = \frac{4}{7} \#$$

(b) 3 1 1 1

$$\Omega = 7 \times 6 \times 5 = 210$$

$$W = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 6$$

$$b = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 24$$

$$\therefore \frac{6 + 24}{210} = \frac{\cancel{30}^{10} \cancel{1}^1}{\cancel{210}^{70} \cancel{7}^7} = \frac{1}{7} \#$$

(c) 3 1 1 1

$$\Omega = \begin{pmatrix} 7 \\ 3 \end{pmatrix} = \frac{5040}{144} = 35$$

$$A = \begin{pmatrix} 3 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 3 \end{pmatrix} = 1 + \frac{24}{6} = 1 + 4 = 5$$

$$\therefore \frac{\cancel{35}^5 \cancel{1}^1}{\cancel{35}^7 \cancel{7}^7} = \frac{1}{7} \#$$